Topic 7 — Hypothesis Testing: Z-Tests and t-Tests

Hypothesis Testing: Motivation

Definition

A **statistical hypothesis** is an assertion concerning the probability distribution of one or more random variables.

Example

Tossing a coin: let p be the probability the coin lands heads. We formulate two hypotheses:

$H_0 : p = \frac{1}{2}$ (fair coin, *null hypothesis*), $H_1 : p \neq \frac{1}{2}$ (unfair coin, *alternative hypothesis*).

The task is to decide, from data such as $H, H, T, H, T, T, H, ...$, whether to reject H_0.

Critical Regions and Significance

Definition

The **critical region** of a statistical test is the set of values of the test statistic which lead us to reject H_0. The **acceptance region** is the set of values leading us to fail to reject H_0.

Remark

Some statisticians say a test can "accept" or "reject" H_0; others prefer "reject" or "fail to reject" H_0, avoiding the word "accept".

If H_1 is **two-sided** (e.g. $H_1 : \mu \neq \mu_0$), we use a **two-tailed test**. If H_1 is **one-sided** (e.g. $H_1 : \mu > \mu_0$ or $\mu < \mu_0$), we use a **one-tailed test**.

The probability of a Type I error is denoted α; this is the **significance level** of the test. E.g. $\alpha = 0.05$: "testing H_0 at the 5% level."

An alternative to critical regions: ***p*-values**. We calculate the probability of a value equal to, or more extreme than, the test statistic. For a one-tailed test this is the p-value, compared with α; for a two-tailed test it is $\frac{p\text{-value}}{2}$, compared with $\frac{\alpha}{2}$.

Types of Error

Definition

A **Type I error** occurs if H_0 is rejected when it is true. A **Type II error** occurs if H_0 is not rejected when it is false.

Remark

Strategy (typical order of actions):

1. Determine H_0, H_1, and the significance level.
2. Decide if a one-tail or two-tail test is appropriate.
3. Calculate the test statistic **assuming H_0 is true**.
4. Compare with the critical value(s), or use the p-value approach.
5. Reject or do not reject H_0.

The Mean of n Observations, σ^2 Known

If $X \sim N(\mu, \sigma^2)$, then $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$. Given a sample $x_1, \ldots, x_n$, we calculate $\bar{x}$ and construct the test statistic

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}},$$

then compare with the critical value(s) or use the p-value approach.

Example: Machine Calibration (Z-test)

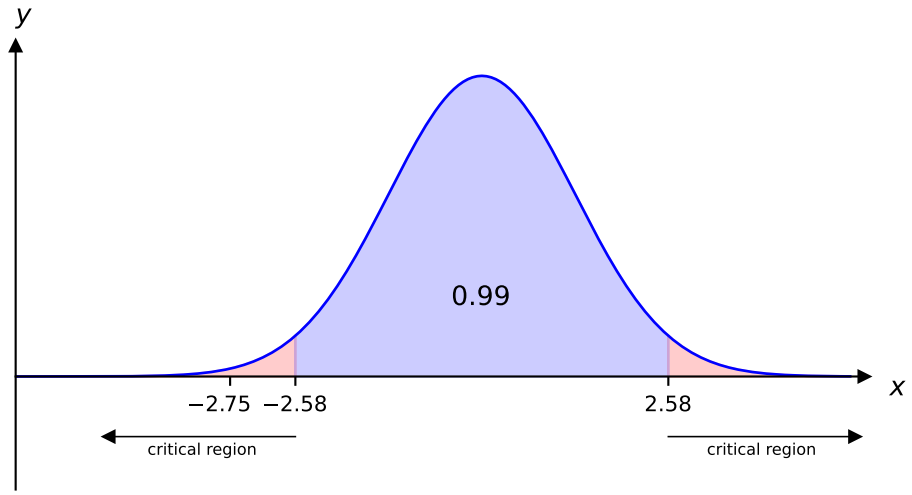
Example

A machine produces items of nominal mass 1kg; $X \sim N(\mu, (0.02)^2)$. If $\mu \neq 1$ the machine needs correcting. The mean of 25 randomly selected items was 0.989kg. Test $H_0 : \mu = 1$ at the 1% level.

$$H_0 : \mu = 1, \qquad H_1 : \mu \neq 1, \qquad \alpha = 0.01.$$

Since $\mathbb{P}(Z \leq 2.58) = 0.995$, the critical region is $|z| > 2.58$, and

$$z = \frac{0.989 - 1}{0.02/5} = -2.75.$$



Example

Since -2.75 is in the critical region, we reject H_0 at the 1% level: the machine should be corrected.
p-value approach: $\mathbb{P}(Z \geq -2.75) = 0.003 < 0.005$ $(\frac{\alpha}{2})$, so we reject H_0.

General Strategy for p-Values / Critical Values

Remark

1. Calculate the statistic (z, t, …).
2a. *p-value approach:* if the statistic > 0, calculate $1 - F(\text{stat})$; if < 0, calculate $F(\text{stat})$; if two-tailed, double the result. Compare p with α: $p < \alpha \Rightarrow$ reject H_0; $p \geq \alpha \Rightarrow$ do not reject.
2b. *Critical-value approach:* find the critical value z_{cr} (or t_{cr}) for α (or $\frac{\alpha}{2}$, two-tailed) from a table/Python, and compare $|z|$ with z_{cr}: $|z| > z_{cr} \Rightarrow$ reject H_0.

The Difference Between Two Means

For samples of size n_1, n_2 from $N(\mu_1, \sigma_1^2)$, $N(\mu_2, \sigma_2^2)$ (variances known), $(\bar{x}_1 - \bar{x}_2) \sim N(\mu_1 - \mu_2,\ \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})$, giving the test statistic

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}.$$

Example: Comparing Two Means

Example

Samples of size 25 from $X \sim N(\mu_1, 66)$ and $Y \sim N(\mu_2, 66)$ give $\bar{x} = 116$, $\bar{y} = 109$. Test at the 5% level:

$$H_0 : \mu_1 - \mu_2 = 12, \qquad H_1 : \mu_1 - \mu_2 \neq 12.$$

Critical region $|z| > 1.96$. Test statistic:

$$z = \frac{(116 - 109) - 12}{\sqrt{2 \cdot \frac{66}{25}}} = -2.17.$$

Since $|-2.17| > 1.96$, reject H_0.

Large Sample Tests

The CLT lets us test hypotheses about means of **non-normal** distributions, provided the sample is large (30+): $\bar{X} \approx N(\mu, \frac{\sigma^2}{n})$, and the test statistic is only *approximately* $N(0, 1)$.

Example: A Golfer's Improvement

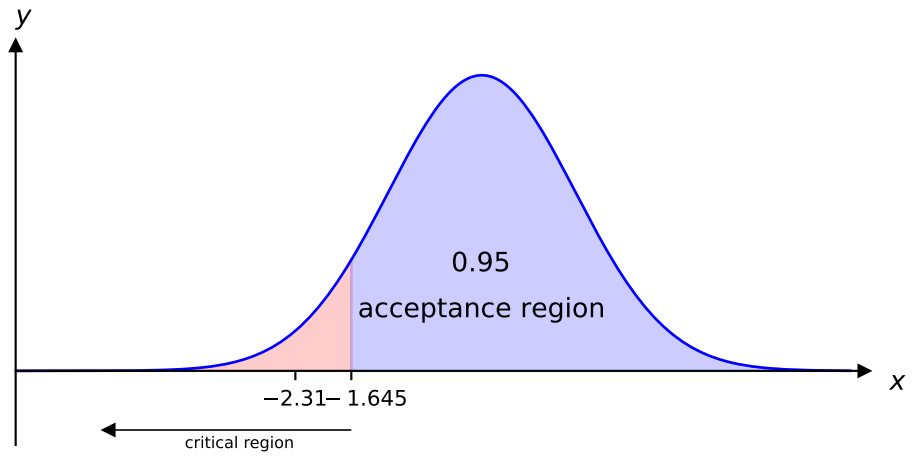
Example

A golfer's round has mean 84.1, sd 2.6. After lessons, mean is 83.1 over 36 rounds. Test at 5%:

$$H_0 : \mu = 84.1, \qquad H_1 : \mu < 84.1 \text{ (one-tailed)}, \qquad z_{cr} = 1.645.$$

$$z = \frac{83.1 - 84.1}{2.6/6} = -2.31.$$

Since $|z| = 2.31 > 1.645$, reject H_0: her game has improved.



p-value approach: $\mathbb{P}(Z \leq -2.31) = 0.01044 < 0.05$, so we reject H_0.

The t-Distribution

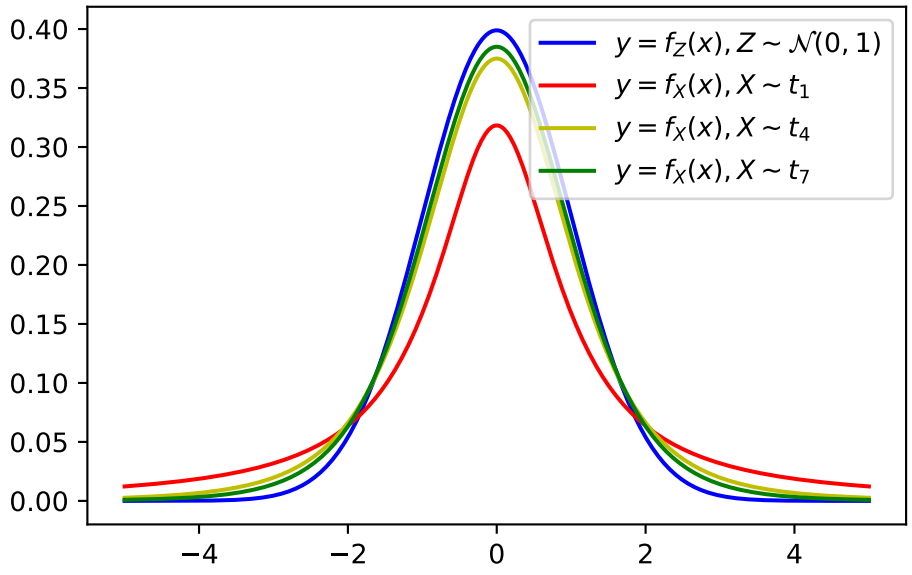
Definition

If $x_1, \ldots, x_n$ is a sample ($n \geq 2$) from $X \sim N(\mu, \sigma^2)$, the random variable

$$T = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

has a **t-distribution** with $(n-1)$ degrees of freedom.

Degrees of freedom = number of values free to vary. E.g. for 4 numbers a, b, c, d with mean 5 (so $a+b+c+d = 20$): once 3 are known, the fourth is determined — only $3 = 4-1 = n-1$ are "free". As $n \to \infty$, the t-distribution tends to the normal; it has heavier tails for small degrees of freedom.



t-Test: Comparing a Sample Mean

For $X \sim N(\mu, \sigma^2)$ with σ^2 **unknown**, testing $H_0 : \mu = \mu_0$: we estimate σ^2 by the sample variance s^2 and use

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim t_{n-1}.$$

Example: Yarn Breaking Strength

Example

Yarn breaking strength has mean 21N. A treated sample of 9 lengths has mean 22.75, sd 2.109. Test at 5%:

$$H_0 : \mu = 21, \qquad H_1 : \mu > 21 \text{ (one-tailed)}.$$

Critical region: $t > t_{0.05,8} = 1.860$.

$$t = \frac{22.75 - 21}{2.109/\sqrt{9}} = 2.5.$$

Since $2.5 > 1.860$, reject H_0: the claim is supported at the 5% level.

Paired t-Test

Used for "before"/"after" data on the **same** items. With n pairs and difference $D = X_1 - X_2 \sim N(\mu_1 - \mu_2, \sigma^2)$, testing $H_0 : \mathbb{E}(D) = \mu_0$:

$$t = \frac{\bar{D} - \mu_0}{s_D/\sqrt{n}} \sim t_{n-1}.$$

Example: Meat Spoilage (Paired t-Test)

Example

Ten joints of meat, each half frozen by process A or B; days to spoilage:

Joint	1	2	3	4	5	6	7	8	9	10
A	63	109	82	156	161	155	47	141	92	149
B	129	105	76	207	253	146	62	160	90	177

$H_0 : \mu_D = 0$, $H_1 : \mu_D \neq 0$ (two-tailed), $D = B - A$.

Example

Pairs	1	2	3	4	5	6	7	8	9	10	Total	Mean
D	66	-4	-6	51	92	-9	15	19	-2	28	250	25
$D - \bar{D}$	41	-29	-31	26	67	-34	-10	-6	-27	3	0	
$(D - \bar{D})^2$	1681	841	961	676	4489	1156	100	36	729	9	10678	

$$s_D^2 = \frac{10678}{9} = 1186.44, \qquad t = \frac{25 - 0}{\sqrt{1186.44/10}} = 2.3.$$

Critical region: $|t| > t_{0.025,9} = 2.26$. Since $2.3 > 2.26$, reject H_0: there is a difference between the processes.

Unpaired t-Test

For two **independent** samples $X_1 \sim N(\mu_1, \sigma^2)$, $X_2 \sim N(\mu_2, \sigma^2)$ (same unknown variance), we pool the variance estimate:

$$S^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{(n_1 - 1) + (n_2 - 1)} \quad \left(= \tfrac{S_1^2 + S_2^2}{2} \text{ if } n_1 = n_2\right).$$

Test statistic ($n_1 + n_2 - 2$ degrees of freedom):

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{S\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}.$$

Example: Energy Drink Trial (Unpaired t-Test)

Example

8 people given water, 8 given an energy drink, then an endurance task:

	Mean	Std
Water (x_1)	12.2	2.4
Energy drink (x_2)	13.1	3.1

$H_0 : \mu_1 - \mu_2 = 0$, $H_1 : \mu_1 < \mu_2$ (one-tailed).

Example

Pooled sd: $S = \sqrt{\frac{2.4^2+3.1^2}{2}} = 2.7722$.

$$t = \frac{(12.2 - 13.1) - 0}{2.7722\sqrt{2/8}} = -0.6493, \qquad n_1 + n_2 - 2 = 14 \text{ d.f.}, \qquad t_{0.05,14} = 1.761.$$

Since $|-0.6493| < 1.761$, we do **not** reject H_0: no evidence the energy drink improves performance.